Beyond Centroids: A Density-Based Exploration of Non-Linear Clustering using DBSCAN on Spatial Data

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GitHub: https://github.com/Keerthi99-v/machine-learning-

Introduction:

Clustering is a fundamental technique used to identify hidden structures in dataset of various sorts. Conventional clustering techniques like K-Means rely on distance-based partitioning under the convex and well-separated assumption. Nonetheless, many real-world datasets display irregular nonlinear patterns which cannot fit under these assumptions.

In the analysis of spatial data, clusters usually have complex shapes and contain noise/outliers. More flexible approaches capable of adapting to the underlying structure of the data are needed by these features. DBSCAN (Density-Based Spatial Clustering of Applications with Noise) overcomes these limitations by clustering based on local density instead of distance to centroids.

In this tutorial, we shall apply DBSCAN to spatial dataset having non-linear structure. The research study performs parameter selection using the k-distance graph, assesses the clustering behaviour of DBSCAN under several conditions, and contrasts DBSCAN and K-Means to show they are different. The goal is to show that density-based clustering gives a more realistic and interpretable representation of complex data.

Theoretical Background:

Clustering is an unsupervised learning task that groups data points based on similarity. K-Means clustering minimises within-cluster variance but assumes that clusters are spherical and requires the number of clusters to be predefined.

DBSCAN, in contrast, defines clusters as dense regions separated by sparse areas. It uses two parameters: eps, which defines the neighbourhood radius, and min_samples, which specifies the minimum number of points required to form a cluster.

Points are categorised as core points, border points, or noise. This allows DBSCAN to detect outliers and capture clusters of arbitrary shape.

A key challenge in DBSCAN is selecting appropriate parameters. The k-distance graph provides a visual method for identifying a suitable value of eps by highlighting a transition between dense and sparse regions.

Methodology:

A synthetic spatial dataset with non-linear structure was generated to evaluate clustering performance. The dataset contains two interleaving curved shapes, making it ideal for testing density-based methods.

DBSCAN was applied with different parameter settings to analyse clustering behaviour. The k-distance graph was used to guide parameter selection. Additionally, K-Means clustering was applied to the same dataset for comparison.

Results and Visual Analysis

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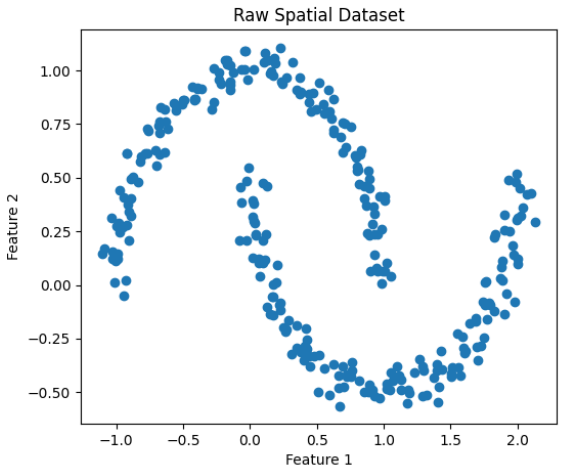
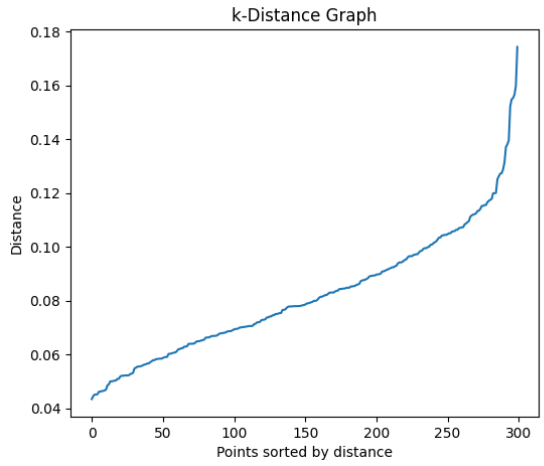


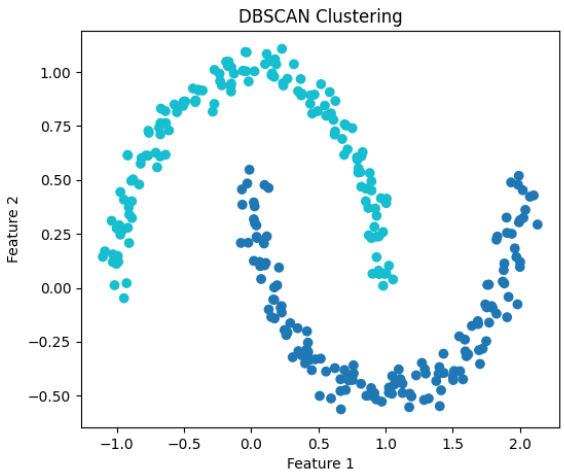
The dataset exhibits two curved, interleaving shapes that cannot be separated using linear boundaries. This structure highlights the limitations of traditional clustering methods and motivates the use of DBSCAN.

K- Distance Graph:



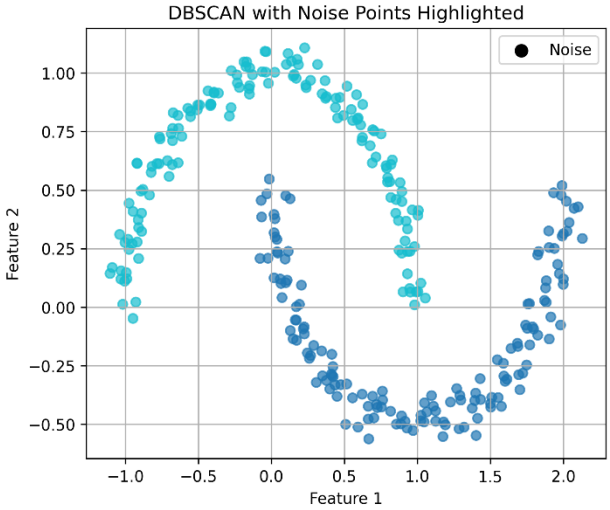
The k-distance graph shows a noticeable change in slope, indicating the transition between dense regions and noise. Selecting *eps* near this point ensures that clusters are correctly identified while avoiding excessive noise.

DBSCAN Clustering:



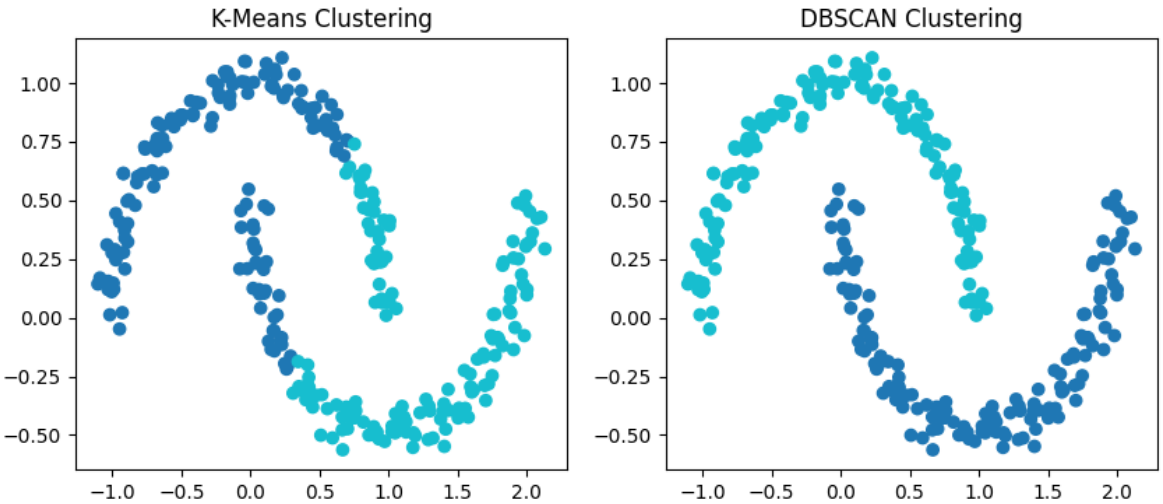
DBSCAN successfully identifies the two curved clusters, following the natural shape of the dataset. This demonstrates its ability to capture non-linear structures.

Noise Detection:



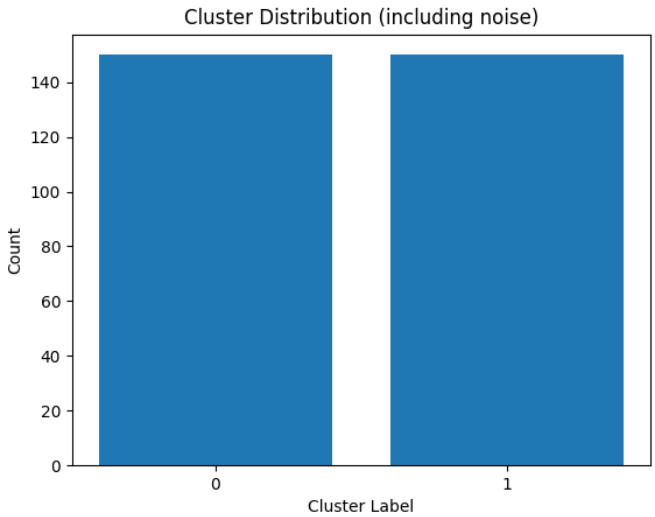
Noise points are identified as observations that do not belong to any dense region. The small number of noise points indicates that the dataset is clean and well-structured.

Comparison with K – Means:



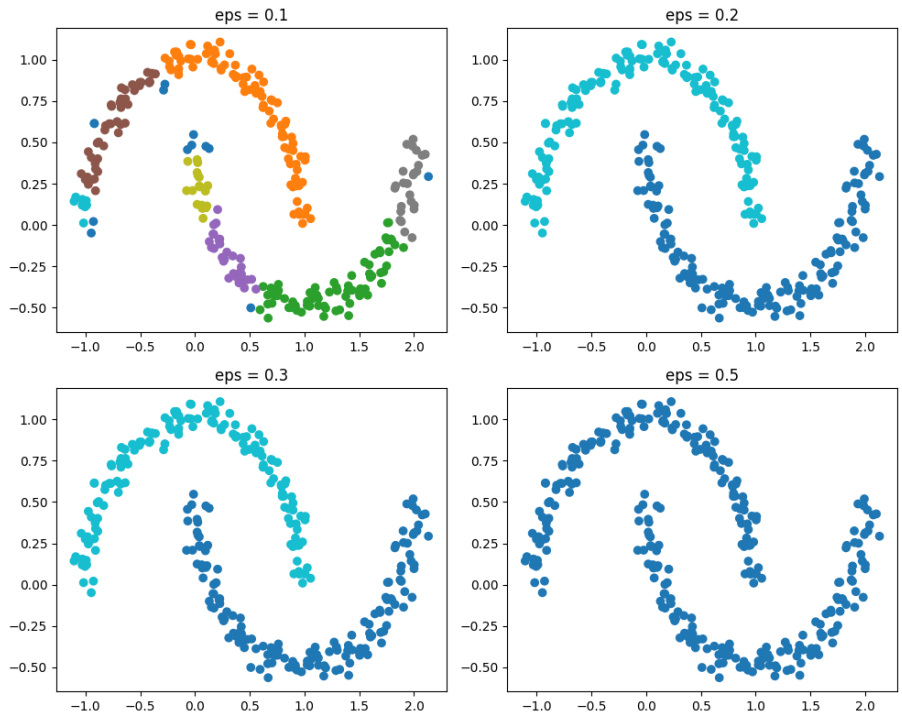
K-Means fails to capture the curved structure and instead divides the dataset into linear clusters. DBSCAN, however, follows the natural shape of the data, highlighting its superiority for non-linear clustering.

Cluster Distribution:



The distribution plot shows the number of points in each cluster, including noise. This provides insight into cluster density and the proportion of outliers.

Parameter Sensitivity:



Different values of eps produce varying clustering results. Smaller values result in more noise, while larger values merge clusters. This demonstrates the importance of careful parameter tuning.

Discussion:

The findings of this study show that the DBSCAN clustering algorithm can detect spatial data that have a non-convex shape. DBSCAN unlike centroid-based approaches imposes restrictions on cluster shapes geometrically. The clustering results provide convincing evidence for this claim, as we see that the algorithm captures the curved “moon-like” shapes in the data. The K-Means method is ineffective because such shapes cannot be accurately defined using linear boundaries.

DBSCAN can separate out noise points present in the dataset supplied. The analysis showed that very few points were labelled noise. This indicates that the data is clean and well-structured. Nonetheless, the existence of these noise points is still significant because they show that DBSCAN does not require every observation to belong to a cluster. This behaviour is especially useful in real-world settings where outliers may refer to anomalies or rare occurrences.

The k-distance graph is helpful for choosing parameters. Instead of randomly choosing a value for eps, the graph shows us the proper data to select one where after that point the distances start to increase rapidly. This transition is the line separating dense groups from sparse groups. By selecting eps near the elbow point, we ensure that clusters are not split-up and not merged too much.

DBSCAN has more advantages than K-Means as shown in the comparison. K-Means fails to represent the dataset as a linear segment as it does not have the structure of the dataset. Thanks to its reliance on distance to centroids in a convex cluster, K-Means works well. On the other hand, DBSCAN is dependent on the distribution of the dataset and creates clusters that reflect the true shape of the database. Choosing clustering algorithms based on data properties rather than for convenience is a major highlight in finding.

The effectiveness of DBSCAN is reliant on the parameter of eps depending on analysis. As the value gets smaller, more noise is detected and more fragmented clusters are found. As the value gets larger, clusters are combined that are distinct. DBSCAN is powerful, but care and tuning must be exercised in its use. Parameter variation plots bolster the analysis by demonstrating how clustering behaviour shifts across various settings.

The results suggest that DBSCAN is ideal for spatial data with an irregular structure and noise. It is a suitable alternative for classic clustering methods due to its flexibility and its ability to identify outliers while avoiding fixed number clusters. Nonetheless, successful utilization of this method relies on careful parameter selection and knowledge of the data distribution.

Conclusion:

This study demonstrates that DBSCAN is a powerful clustering technique for non-linear data.

- It successfully identifies clusters of arbitrary shape.
- It detects noise and outliers effectively.
- The k-distance graph supports parameter selection.
- DBSCAN outperforms K-Means for complex datasets.
- Parameter tuning is essential for optimal results.

Future Work:

- Apply DBSCAN to real-world geospatial datasets.
- Explore adaptive parameter tuning methods.
- Combine DBSCAN with dimensionality reduction.
- Extend analysis to higher-dimensional datasets.

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